

Addis Ababa University
College of Natural Sciences
Department of computer science
Web Application Course outline

1. Course objectives

Upon the completion of this course, trainers will be able to:

- understand how websites are stored and served over the Internet
- develop websites with HTML, CSS and JavaScript
- develop a database-driven, dynamic web application with PHP and MySQL
- maintain, update, and manage websites

2. Course outline

1. Introduction

- The Internet
- The World Wide Web (WWW)
- The Client/Server Architecture of the World Wide Web
- The Domain Name System (DNS)
- URL, URI, URN
- HTTP

2. Static HTML/XHTML

- XHTML Document Structure and XHTML Tag Format
- Core XHTML tags
- Presentation and Layout
- Interactivity
- Cascading Style Sheets (CSS)
 - Basic CSS
 - More CSS
 - Styling Page Sections
 - Introduction to Layout
 - Floating Elements
 - Sizing and Positioning
- Design issues

3. Client-Side Scripts

- Client-Side Scripting (with JavaScript)
- Dynamic HTML (DHTML)
- The Document Object Model (DOM)
- Application of Client-Side Scripts
- Browser Compatibility Issues

4. Server-Side Programs

- The Common Gateway Interface (CGI)
- Server-Side Scripting with PHP
 - PHP Basic Syntax
 - Embedded PHP
 - Advanced PHP Syntax
 - Parameterized Pages

- Processing Form Data in PHP
- Object-Oriented PHP
- Session Management
- Database Driven Web Applications

5. Some Advanced Concepts

- The eXtensible Markup Language (XML)
- Java Servlets & Java Server Pages (JSP)

Reference books:

- Thomas A. Powell, *HTML & XHTML: The Complete Reference* (McGraw-Hill)
- Neil Gray, *Web Server Programming* (John Wiley & Sons)
- Rasmus Lerdorf, Peter MacIntyre, Kevin Tatroe, *Programming PHP*, 2nd Edition (O'Reilly)
- Goodman, *Java script bible* 3rd ed. (Gage Publishing)
- Gundavaram, S. *CGI Programming on the World Wide Web* (O'Reilly and Associates Publishing)
- Horton, William, *The Web Page Design Cookbook*. (John Wiley & Sons)